

Predictive Maintenance

Technical and Executive Brief

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This document presents the predictive maintenance model from both an engineering and business perspective. It outlines the model architecture, sensor features, performance evaluation, financial impact, and recommended implementation steps.

1. Technical Brief

To: Engineering Team

From: Data Science Team

Subject: Predictive Maintenance Model Architecture, Metrics and Sensor Logic

Model Overview

A Logistic Regression classifier has been developed to predict bearing failure from real time vibration measurements. The system processes 500 samples per minute and produces a probability of failure. This probability is evaluated against an alert threshold to support maintenance decisions.

Sensor Feature Engineering

Feature	Formula	Purpose
Rolling Mean (SMA)	$\mu = \Sigma(x_{\text{roll}}) / n$ over 20 samples	Tracks the baseline degradation trend and reduces noise.
Rolling RMS	$RMS = \sqrt{(\Sigma(x_{\text{roll}}^2) / n)}$	Measures vibration energy and increases as bearing wear develops.
Rolling Range	$Max(x_{\text{roll}}) - Min(x_{\text{roll}})$	Captures peak severity and short duration vibration spikes.

Engineering rationale: Rolling RMS provides a strong indicator of increasing vibration energy associated with bearing wear. Rolling Range complements this by identifying transient impacts that may not be visible in the average signal.

Model Performance

Metric	Value	Interpretation
Accuracy	[Insert validated result]	Overall classification accuracy.
Sensitivity (Recall)	[Insert validated result]	Percentage of actual failures detected.
Specificity	[Insert validated result]	Percentage of healthy states correctly identified.
Precision	[Insert validated result]	Percentage of failure alerts that were genuine.
F1 Score	[Insert validated result]	Balance between precision and recall.

	Predicted Healthy	Predicted Failing
Actual Healthy	TN	FP
Actual Failing	FN	TP

Failure management priority: False negatives are the highest risk because a missed failure can result in unplanned downtime. False positives may result in additional inspection or maintenance, making threshold selection an operational trade off.

Deployment and Threshold Strategy

- **Initial alert threshold:** $P(\text{Failure}) > 0.6$.
- The threshold can be adjusted according to the relative cost of missed failures and unnecessary maintenance.
- **Retraining:** Review and retrain the model approximately every 30 days as new run to failure data becomes available.

Recommended Technical Next Steps

- Deploy the model to an edge device such as a Raspberry Pi or PLC for real time inference.
- Integrate temperature and pressure measurements to improve the feature set.
- Create an operational dashboard displaying Rolling RMS, Rolling Range and Failure Probability.

2. Executive Brief

To: Chief Financial Officer

From: Data Science Team

Subject: Predictive Maintenance: ROI, Cost Savings and Risk Reduction

Business Problem

The current maintenance approach combines reactive repairs with scheduled overhauls. This creates avoidable costs through emergency repairs, production interruptions and maintenance performed before it is actually required.

- Unplanned downtime can exceed \$50,000 per event when lost production and emergency repair costs are combined.
- An estimated 30 to 40 percent of scheduled maintenance activity may represent avoidable expenditure.

The Proposed Solution

The predictive maintenance system monitors equipment vibration and identifies patterns associated with impending bearing failure. Instead of waiting for equipment to fail, maintenance teams receive an early warning that allows work to be planned before an unplanned outage occurs.

Current demonstrated capability: Bearing failure can be predicted up to **130 minutes in advance**, providing an opportunity to coordinate maintenance during a planned shift change.

Estimated Financial Impact

Cost Category	Current Annual Cost	Projected with PdM	Annual Savings
Emergency Repairs	\$250,000	\$75,000	\$175,000
Unplanned Downtime	\$500,000	\$200,000	\$300,000
Scheduled Maintenance	\$150,000	\$90,000	\$60,000
Total	\$900,000	\$365,000	\$535,000

Projection assumption: 80 percent of failures are identified early and 70 percent of emergency events are avoided.

Return on Investment

Implementation Cost	~\$50,000
Projected Annual Savings	~\$535,000
Year 1 ROI	970%
Estimated Payback	Less than 1 month

Risk Reduction

- **Equipment risk:** Early warning reduces exposure to catastrophic failure scenarios.
- **Production risk:** Planned interventions can reduce the frequency and duration of unplanned outages.
- **Safety risk:** Moving from emergency repairs toward planned maintenance reduces exposure to high pressure repair situations.

Recommendation

Approve an initial investment of \$50,000 for sensor deployment and model integration. Under the stated assumptions, the project has the potential to recover its implementation cost within weeks and deliver approximately \$535,000 in annual savings.

3. Reflection

Switching Between Technical and Executive Contexts

Switching between the technical and executive briefs was both challenging and enlightening.

The technical brief was more natural because it allowed me to communicate directly using concepts such as RMS, sensitivity, confusion matrices and feature engineering. These terms are useful when the audience needs to understand how the predictive maintenance system works.

The executive brief required a different approach. Technical achievements had to be translated into financial value, operational risk and measurable business outcomes. Instead of focusing on the mechanics of logistic regression or rolling windows, the emphasis shifted toward avoided downtime, cost savings, risk reduction and return on investment.

The exercise reinforced an important lesson: the value of data science is not limited to model performance. A technically strong model only creates business value when its results can be understood, trusted and acted upon by the people responsible for operational and financial decisions.

Going forward, I will focus on communicating business outcomes first and providing technical detail according to the needs of the audience. This approach will allow me to communicate more effectively with both technical teams and business leadership.

Data validation note: Model performance values should be populated from the final validated evaluation output before the technical brief is used for formal model approval.